

DSA 210

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This project aims to analyze traffic accidents in Turkey between 2020 and 2024, focusing on the impact of weather conditions and time of day on accident severity.

Traffic accident data will be collected from TÜİK, while weather data will be obtained from the Open-Meteo Historical Weather API. The dataset will include monthly accident counts, fatal and injury rates, average rain levels across the country by months, and time-related information.

The study is based on two main hypotheses. First, the rate of fatal accidents increases during rainy weather conditions across the country. Second, accidents occurring at night have a higher fatality rate compared to those occurring during the day.

To measure the impact of weather conditions, Turkey's monthly accident statistics from TÜİK will be used and compared with the average rain levels of months in Turkey. To analyze night-time accidents, accident reports categorized by daylight conditions (day/night) from TÜİK will be examined to evaluate differences in fatality rates.

In the later stages of the project, the study aims to extend the analysis by building simple predictive models. Using weather conditions and time-of-day information, the project will attempt to estimate the likelihood of accidents and predict whether an accident is more likely to result in a fatality or injury. These predictions will provide additional insights into risk factors and contribute to a better understanding of accident severity patterns.